Assignment 1 - Functions for Spatial Computing

Submission Date: Tuesday, September 8th

This assignment utilizes concepts introduced in the Week 2 notebook focusing on functions. You can use built-in functions and create reusable user-defined functions for introductory spatial problems involving distance, density, area, and geographic coordinates (key components for spatial computing).

Complete the work in a new Google Colab notebook. Put your name, course number, and assignment title at the top; label every part clearly; run every code cell; and leave all output visible. Use descriptive snake_case names and include units where appropriate.

Table 1 outlines the required assignment components.

Component	Requirements	Points
1. Built-in function warm-up	For park_areas = [12.75, 8.4, 21.6, 15.25, 10.0], use built-in functions to display the count, sum, minimum, maximum, rounded sum, and a new ascending list. Do not manually type calculated answers.	10
2. Convert distance units	Define a function called miles_to_kilometers(miles), using 1 mile = 1.60934 kilometers. Return rather than print the result. Test 1, 5, and 12.5 miles and display each result to two decimal places.	15
3. Calculate population density	Define a function called population_density(population, area_sq_miles) and return people per square mile. Test North District (48,500; 12.4 sq mi), Central District (76,200; 8.7 sq mi), and South District (35,900; 20.1 sq mi). Display whole-number results.	15
4. Use a default parameter	Define a function called buffer_area(radius, pi_value=3.14159) and return pi_value * radius ** 2. Call it for a 500-meter radius using both the default and pi_value=3.14. Display and briefly explain the differing results.	15
5. Work with coordinates	Define a function called coordinate_midpoint(x1, y1, x2, y2). Calculate the coordinate halfway between two points using: midpoint_x = (x1 + x2) / 2 midpoint_y = (y1 + y2) / 2 Return both midpoint coordinates. Test the function using (10, 20) and (30, 40), then (5.5, 12.0) and (18.5, 24.0). Store the two returned values using variable unpacking and print each midpoint with a descriptive label.	15
6. Document a spatial function	Define a function called rectangle_summary(length, width, unit="meters") which calculates the rectangle's area and perimeter. Include a docstring describing the function, parameters, returned values, and an example call. Include at least two useful inline comments. Test the function with (i) a study area measuring 120 by 75 meters, and (ii) a study area measuring 0.8 by 0.5 kilometers, overriding the default unit.	20
7. Design your own spatial function	Create a new function for a simple spatial or geographic problem. It must have a descriptive name, at least two parameters, a meaningful calculation or transformation, a return value, a concise docstring, and three tests. Explain its purpose and usefulness in two or three sentences.	10

Table 1 Assignment components and requirements.

Submit the completed .ipynb file online. Before submitting, restart the runtime and run the notebook from top to bottom. Confirm that every function is called and tested, requested docstrings and written explanations are present, and no unexpected errors remain.

Also submit a .PDF file of your notebook (or a .PDF of screenshots), for easy review.

Table 2 outlines the assessment criteria.

Criterion	Evidence	Weight
Correctness and completeness	All seven parts are present, across calculations, conditions, returned values, and the final displayed results are correct.	40%
Function design	Functions use appropriate parameters, arguments, default values, return statements, and reusable logic rather than hard coded answers.	25%
Documentation and readability	Names preferably follow snake_case, units and labels are clear, docstrings are present, and the comments and written explanations are useful.	15%
Testing and output	Every function is tested with the required inputs. Cells have been run and outputs provide evidence that the code works.	15%
Organization and submission	The notebook is clearly labeled, logically ordered, complete, and submitted in the required format.	5%

Table 2 Assessment criteria.

On this particular assignment, you are not allowed to use AI, otherwise it defeats the point of trying to learn this content. The knowledge from this assignment is cumulative and necessary to succeed in later weeks of the course (and especially in the pencil and paper exam where you cannot use AI).

Undisclosed AI-generated work, fabricated material, or code that the student cannot explain may constitute an Honor Code violation.